BrewGuide

Daily Suggestion · Feature Elaboration

The Daily Suggestion surfaces one carefully chosen brewing technique each day — deterministic enough to feel shared and social, personalised enough to feel curated. This document covers the selection algorithm, UI card spec, caching strategy, notification hook, data model changes, and the personalisation signals available from the existing offline-first brew journal.

1. Core concept

The strongest implementation uses a two-layer system: a **global daily seed** that every user shares (so the same technique appears for everyone on a given date, enabling social conversation), layered with a **personal weighting pass** that re-ranks candidates based on brew history. Pure personalisation creates filter bubbles; pure randomness feels arbitrary. This hybrid avoids both.

Design principle

Global seed → shared social moment, same pick worldwide per day

Personal filter → skip recently brewed, boost neglected categories

Deterministic → recomputes to the same answer on the same calendar day

Offline-first → zero network calls, all signals from CoreData brew journal

2. Selection algorithm

The pipeline runs in five stages each time the suggestion is recomputed (once per calendar day, or on first launch if the cache is stale):

1 Daily seed	2 Top 5 window	3 Recency filter	4 Score & rank	5 Pick winner
daysSinceEpoch % totalTechniques Same result globally per day	Rotating candidate window around today's seed index	Remove techniques brewed in the last 14 days	Difficulty + diversity + seasonal signal weighting	Highest scorer wins Fallback: global pick

Algorithm stages — left to right: seed → candidate window → recency filter → score → winner

- **Stage 1 — Daily seed.** Compute `daysSinceEpoch % totalTechniques`. This index is stable for the entire calendar day and identical for every user.
- **Stage 2 — Candidate window.** Take the 5 techniques nearest to the seed index (wrapping around the array). This gives variety without pure randomness.
- **Stage 3 — Recency filter.** Query CoreData for techniques brewed in the last 14 days and remove them from candidates. If all 5 are filtered, skip to the global pick (fallback).
- **Stage 4 — Score and rank.** Apply the weighted scoring table (Section 3) to each remaining candidate. Higher score wins.

- **Stage 5 — Cache result.** Store the winning technique ID and today's date in UserDefaults. Serve the cached result until midnight.

3. Scoring signals

All signals are computable from the local CoreData BrewSession store — no server required. Weights are suggestions; tune after observing real usage.

Signal	Weight	Notes
Difficulty match	**+30 pts**	Boost if technique difficulty matches user's most-brewed tier
Method diversity	**+25 pts**	Penalise if same method category as yesterday's brew
Seasonal fit	**+20 pts**	data-driven per-technique seasonalWeight[season] multiplier
Not tried before	**+15 pts**	Boost techniques never attempted by this user
High journal rating	**+10 pts**	If tried before, boost if previous rating was 4★ or 5★
Brewed recently	**–40 pts**	Heavy penalty if brewed within 14 days
Poor rating	**–20 pts**	Suppress if previously rated 1★ or 2★

4. Home screen card specification

The card lives at the top of the home screen above the method category carousel — the editorial position. It needs one compelling reason to tap, not a full brief.

Element	Detail
TODAY badge	Date string (e.g. 'Mon 22 Mar') with a small gold dot
Technique name	Large, prominent — the hero element of the card
Equipment name	Smaller, secondary, below the technique name
Flavour hook	Single evocative line from the database — e.g. 'Bright & floral'
Brew time	Estimated total time (e.g. '~4 min')
Difficulty	3 filled / 2 empty dots — never a raw number
Previous brew	If brewed before: 'You last made this — 4★' in muted text
Brew now CTA	Full-width button, deep-links to technique with params pre-filled

5. Caching and midnight refresh

The suggestion must not change during an active session and must update at midnight local time — not UTC, not on next launch.

- Cache the selected technique ID + calendar date in `UserDefaults`.
- On each app foreground event, compare today's date to the cached date. Recompute if different.
- Subscribe to `UIApplication.significantTimeChangeNotification` to catch midnight transitions while the app is open.
- This means a user who leaves the app open overnight sees the new suggestion appear automatically.

Swift data structure

```
struct DailySuggestion {

let technique: Technique

let computedDate: Date

let personalScore: Double // 0–1, for debugging

let reasonLabel: String // e.g. 'You haven't tried pour over in 3 weeks'

let streakDays: Int // 0 if no active streak

let hasPreviousBrew: Bool

let previousBestRating: Int? // 1–5 if brewed before

}
```

6. Daily notification (opt-in retention hook)

A daily push notification — off by default, opt-in from Settings — is the highest-leverage retention mechanic you can add at zero cost once the algorithm exists. Implementation: schedule a `UNCalendarNotificationTrigger` for 8am daily. Update the trigger each time the daily suggestion recomputes. Deep-link payload routes directly to the suggestion card.

Example notification

Title: "Today's brew — James Hoffmann V60"

Body: "Bright & floral · ~4 min · Try something new today"

Action: Deep-link brewguide://suggestion

7. JSON database changes

One new field per technique in the existing JSON database enables seasonal weighting. It defaults to `[1.0, 1.0, 1.0, 1.0]` if absent (backward-compatible).

Field / Example	Value
seasonalWeight	[spring, summer, autumn, winter] — four floats, default 1.0 each

Cold Brew	[0.6, 1.4, 0.8, 0.4] — strong summer bias
Siphon	[1.0, 0.8, 1.1, 1.3] — winter boost
AeroPress	[1.0, 1.0, 1.0, 1.0] — all-season, no adjustment

8. Personalisation signals (from CoreData)

All signals are derivable from the existing `BrewSession` CoreData entity — no new schema required.

Signal	Usage
Most-brewed category	Boost techniques from underexplored categories
Average rating	Suppress techniques rated 1–2★; boost 4–5★
Category gap	Surface neglected methods after 30+ days without
Serving size preference	Promote single-serve techniques to solo brewers
Brew time of day	Morning brewers may prefer faster techniques
Total brews	Beginners get gentler difficulty; veterans get stretch picks

9. Streak mechanic

A streak counter — 'You've tried the daily suggestion 4 days in a row' — adds habit-formation with minimal implementation. Store the last date the user tapped 'Brew now' from the suggestion card. Compute the streak on launch. Show it as a secondary line on the card ('4-day streak') when active. Reset on a missed day.

10. Implementation checklist

- DailySuggestionService created — seed, filter, score, cache pipeline
- seasonalWeight field added to technique JSON (with fallback default)
- DailySuggestion struct defined with all fields including reasonLabel
- Home screen card component built — above method carousel
- Card shows previous brew note if BrewSession history exists
- Brew now CTA deep-links to technique with params pre-filled
- UserDefaults cache keyed to calendar date (local timezone, not UTC)
- significantTimeChangeNotification listener wired for midnight refresh
- Optional daily notification scheduled via UNCalendarNotificationTrigger
- Notification toggle added to Settings (off by default)
- Streak counter stored and displayed on card when active
- Fallback to global pick when journal is empty or all candidates filtered
- Unit tests covering: seed computation, recency filter, scoring, fallback